Addendum — the face/scene difference is statistically significant

Computed 4 September 2026 by `emoviz/significance.py`. This closes the one inferential gap in the analysis: every subset was previously reported against its own random-pairing null, but the face and scene rates had never been tested **against each other**.

Results

	face-classified	scene-classified	difference
Valence-sign agreement	73.9% [68.2, 79.2]	42.9% [33.3, 51.1]	**+30.9% [+22.6, +40.6]**
Emotion-family agreement	61.9% [55.7, 68.0]	35.6% [26.7, 44.1]	**+26.3% [+18.4, +35.3]**

95% confidence intervals from a clustered bootstrap, 10,000 draws. Clustered permutation test, 10,000 draws: **p = 0.0001** for both metrics.

Why the resampling is clustered

249 of the 1,012 analysed records repeat an earlier prompt, so records are **not independent**. Both procedures therefore resample and shuffle whole **prompt groups**, never individual records. Resampling records would have understated the uncertainty and inflated significance.

What this settles, and what it does not

Settles: the gap between face-classified and scene-classified drawings is not sampling noise. Neither confidence interval for the difference comes close to zero.

Does not settle: the partial circularity. The face/scene split is derived from the same BLIP-VQA output whose description the image reader then reads. A significant difference shows the two groups differ; it does not show that faces inherently carry more emotional signal. That distinction must be stated whenever the result is quoted.

Note also what actually drives the split: 414 of the 421 scene-classified records (98.3%) are ones where the describer reported **no people present** — and 406 of those *do* contain an expression word. The split is governed by the people-count answer, not by absent expression content.

The sentence to use

> The difference is statistically significant — a clustered permutation test gives p = 0.0001, and the 95% confidence interval on the difference is +22.6 to +40.6 percentage points for valence-sign agreement. I clustered on prompt identity because 249 records repeat an earlier prompt. What that establishes is that the gap is not noise; it does not remove the circularity, because the split comes from the same describer output the emotion reader reads.

Replaces

Wherever the documents or the deck say *"no formal significance test of the face–scene difference has been run"*, that is now superseded. The deck's slide 8 footnote and slide 10's "not established" column should be updated when the dissertation is written — but **not tonight**, and the honest version remains true either way, because the circularity caveat still stands.